

Problem A

Basketball

Source file: basketball.{ c | cpp | java | py }

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Little Johnny, our intrepid friend, likes to spend his free time watching basketball practices in the sumptuous gymnasium of the Esbornian capital. While his friends argue about who will score the most baskets, Johnny's programmer spirit makes him notice something different: the trajectory of the ball as it falls to the ground.

He noticed that whenever the ball is dropped from a height H , it hits the ground and bounces. With each new bounce, the ball reaches a height that is a fraction F of the previous height. Johnny realized that the ball travels a considerable distance while bouncing repeatedly.

Your task is to help Johnny calculate the total distance traveled by the ball (going up and down) until the moment it touches the ground for the N -th time.

Input

A real value H ($1.0 \leq H \leq 1000.0$), representing the initial height from which the ball is dropped.

A real value F ($0.0 < F < 1.0$), representing the fraction of the height that the ball recovers after each bounce.

An integer value N ($1 \leq N \leq 100$), representing the bounce number (ground touch) that Johnny wants to observe.

Output

Print a single line containing the phrase "Total distance: " followed by the value of the distance traveled, with two decimal places.

Example of Input 1

10.0 0.5 2

Example of Output 1

Total distance: 20.00

Example of Input 2

5.0 0.2 3

Example of Output 2

Total distance: 7.40

Example of Input 3

100.0 0.1 1

Example of Output 3

Total distance: 100.00
